

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0983

Chatom 138 kV
April 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0983. The Interconnection Customer proposes a 95.5 MW solar photovoltaic facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on April 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$79,342,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Crozet – Rvital Substation 138 KV L
Contingency: Declo Substation – New London Substation 138 KV
Loading: 204.8% (Rating: 167 MVA)
DFAX: 5.4%
Pre-project loading: 201.8%
2. Facility: Detour Substation – Cedarville Junction Substation 138 KV
Contingency: Nichols – St Louis 138 KV
Loading: 190.8% (Rating: 248 MVA)
DFAX: 22.0%
Pre-project loading: 182.3%
3. Facility: Page – Cedarville Junction 138 KV Line
Contingency: Nichols – St Louis 138 KV
Loading: 189.7% (Rating: 252 MVA)
DFAX: 22.0%
Pre-project loading: 181.3%
4. Facility: Purdy – Page 138 KV L
Contingency: St Ouis – Nichols Substation 138 KV
Loading: 187.7% (Rating: 261 MVA)
DFAX: 22.0%
Pre-project loading: 179.6%
5. Facility: Nuclear Fuels – Alvin 138 KV
Contingency: Oak Crk Station – Sans Souci 138 KV
Loading: 181.7% (Rating: 201 MVA)
DFAX: 18.4%
Pre-project loading: 173.0%

6. Facility: Revital – Dale Mabry 138 KV
Contingency: Declo Substation – New London 138 KV L
Loading: 180.0% (Rating: 193 MVA)
DFAX: 5.4%
Pre-project loading: 177.3%
7. Facility: Dale Mabry – Grottoes 138 KV Line
Contingency: New London – Declo 138 KV
Loading: 169.1% (Rating: 215 MVA)
DFAX: 5.4%
Pre-project loading: 166.7%
8. Facility: Detour – Siloam Springs 138 KV Line
Contingency: Detour – Ft Benton 138 KV Line
Loading: 161.7% (Rating: 181 MVA)
DFAX: 14.7%
Pre-project loading: 154.0%
9. Facility: Tes Filer City Sta Substation – Bridge Riv 138 KV line
Contingency: American – Urban 345 KV
Loading: 159.1% (Rating: 223 MVA)
DFAX: 8.2%
Pre-project loading: 155.5%
10. Facility: Elkton D.P – Roosevelt Rd 138 KV
Contingency: Nuclear Fuels – St Louis Substation 138 KV
Loading: 158.3% (Rating: 197 MVA)
DFAX: 12.5%
Pre-project loading: 152.2%
11. Facility: Houston – Elkton D.P 138 KV L
Contingency: St Louis – Nuclear Fuels 138 KV
Loading: 156.7% (Rating: 199 MVA)
DFAX: 12.5%
Pre-project loading: 150.7%
12. Facility: Roosevelt Rd – Wadsworth 138 KV
Contingency: Nuclear Fuels – St Louis Substation 138 KV
Loading: 154.0% (Rating: 208 MVA)
DFAX: 12.5%
Pre-project loading: 148.2%
13. Facility: Mcauley – Crozet 138 KV L
Contingency: Declo – New London 138 KV
Loading: 151.0% (Rating: 211 MVA)
DFAX: 5.4%
Pre-project loading: 148.6%
14. Facility: Cherry – Mount Jackson 138 KV
Contingency: Chatom – Purdy 138 KV line
Loading: 150.6% (Rating: 178 MVA)
DFAX: 7.8%
Pre-project loading: 146.4%
15. Facility: St Louis – Nuclear Fuels 138 KV
Contingency: Purdy – Page 138 KV
Loading: 149.3% (Rating: 192 MVA)
DFAX: 18.0%
Pre-project loading: 140.3%
16. Facility: Dpc Prairie Du Chien – Cherry 138 KV L
Contingency: Purdy – Chatom 138 KV L

- Loading: 144.4% (Rating: 192 MVA)
DFAX: 7.8%
Pre-project loading: 140.5%
17. Facility: Chatom – Alvin 138 KV Line
Contingency: Oak Crk Sta – Sans Souci 138 KV Line
Loading: 143.0% (Rating: 208 MVA)
DFAX: 18.4%
Pre-project loading: 134.5%
18. Facility: Detour – Fort Benton 138 KV
Contingency: Detour – Siloam Springs 138 KV line
Loading: 140.5% (Rating: 250 MVA)
DFAX: 15.8%
Pre-project loading: 134.5%
19. Facility: Mcauley – Video Pd 138 KV L
Contingency: Declo – New London 138 KV
Loading: 128.6% (Rating: 228 MVA)
DFAX: 5.4%
Pre-project loading: 126.3%
20. Facility: St Luis Substation – Nichols 138 KV Line
Contingency: Page – Purdy 138 KV
Loading: 126.7% (Rating: 346 MVA)
DFAX: 20.3%
Pre-project loading: 121.0%
21. Facility: Puddledock Rd – Mount Jackson 138 KV Line
Contingency: Purdy – Chatom 138 KV
Loading: 124.4% (Rating: 206 MVA)
DFAX: 7.8%
Pre-project loading: 120.8%
22. Facility: S Belleville – Nichols Substation 138 KV
Contingency: Purdy – Page 138 KV
Loading: 123.0% (Rating: 352 MVA)
DFAX: 20.3%
Pre-project loading: 117.5%
23. Facility: Siloam Springs – Lewisburg 138 KV line
Contingency: Detour – Ft Benton Substation 138 KV
Loading: 122.0% (Rating: 246 MVA)
DFAX: 6.3%
Pre-project loading: 119.5%
24. Facility: Purdy – Luray Hydro Station 138 KV
Contingency: Purdy – Chatom 138 KV Line
Loading: 119.4% (Rating: 210 MVA)
DFAX: 7.8%
Pre-project loading: 115.9%
25. Facility: Luray Hydro Sta – Puddledock Rd 138 KV
Contingency: Purdy – Chatom 138 KV
Loading: 119.1% (Rating: 210 MVA)
DFAX: 7.8%
Pre-project loading: 115.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crozet – Revital 138 KV	Rebuild line with higher-capacity conductor	\$4,339,000	7% (\$323,000)
Detour – Cedarville Junction 138 KV	Rebuild line with higher-capacity conductor	\$10,888,000	41% (\$4,481,000)
Page – Cedarville Junction 138 KV Line	Rebuild line with higher-capacity conductor	\$12,139,000	41% (\$4,996,000)
Purdy – Page 138 KV L	Rebuild line with higher-capacity conductor	\$2,965,000	41% (\$1,220,000)
Alvin – Nuclear Fuels 138 KV line	Rebuild line with higher-capacity conductor	\$8,523,000	29% (\$2,502,000)
Revital – Dale Mabry 138 KV	Rebuild line with higher-capacity conductor	\$8,167,000	7% (\$607,000)
Dale Mabry – Grottoes 138 KV	Rebuild line with higher-capacity conductor	\$4,790,000	7% (\$356,000)
Siloam Springs – Deour 138 KV	Rebuild line with higher-capacity conductor	\$15,669,000	100% (\$15,669,000)
Bridge Riv – Tes Filer City Sta 138 KV	Rebuild line with higher-capacity conductor	\$6,267,000	43% (\$2,693,000)
Elkton D.P – Roosevelt Rd 138 KV	Rebuild line with higher-capacity conductor	\$1,616,000	19% (\$311,000)
Houston – Elkton D.P 138 KV L	Rebuild line with higher-capacity conductor	\$2,775,000	19% (\$534,000)
Roosevelt Rd – Wadsworth 138 KV line	Rebuild line with higher-capacity conductor	\$4,577,000	19% (\$881,000)
Mcauley – Crozet 138 KV L	Rebuild line with higher-capacity conductor	\$5,229,000	7% (\$389,000)
Cherry – Mount Jackson 138 KV	Rebuild line with higher-capacity conductor	\$6,589,000	100% (\$6,589,000)
St Louis – Nuclear Fuels 138 KV	Rebuild line with higher-capacity conductor	\$8,333,000	43% (\$3,577,000)
Dpc Prairie Du Chien – Cherry 138 KV L	Rebuild line with higher-capacity conductor	\$3,978,000	100% (\$3,978,000)
Chatom – Alvin 138 KV	Rebuild line with higher-capacity conductor	\$4,794,000	29% (\$1,407,000)
Detour – Fort Benton 138 KV	Rebuild line with higher-capacity conductor	\$4,309,000	69% (\$2,983,000)
Mcauley – Video Pd 138 KV line	Rebuild line with higher-capacity conductor	\$1,580,000	7% (\$117,000)
St Louis – Nichols 138 KV	Rebuild line with higher-capacity conductor	\$3,910,000	15% (\$591,000)
Puddledock Rd – Mount Jackson 138 KV Line	Rebuild line with higher-capacity conductor	\$9,405,000	100% (\$9,405,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nichols – S Belleville Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,345,000	15% (\$656,000)
Siloam Springs – Lewisburg 138 KV line	Rebuild line with higher-capacity conductor	\$8,792,000	47% (\$4,160,000)
Luray Hydro Sta – Purdy 138 KV	Rebuild line with higher-capacity conductor	\$4,164,000	100% (\$4,164,000)
Puddledock Rd – Luray Hydro Sta 138 KV L	Rebuild line with higher-capacity conductor	\$6,753,000	100% (\$6,753,000)
	Total allocated to project		\$79,342,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.